

FIFTH SEMESTER DIPLOMA EXAMINATION IN CIVIL ENGINEERING AND QS & CM

STRUCTURAL DESIGN

MODEL QUESTION PAPER

(Maximum marks: 100)

(Time: 3 Hrs)

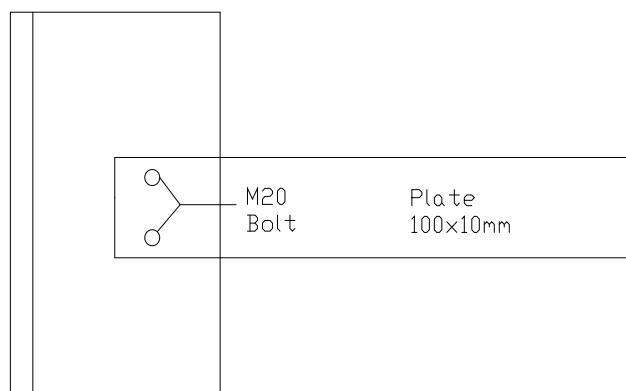
{ Note :- Use of IS 456-2000,IS 800-2007,SP-16 and steel tables are permitted }

PART - A*Answer all questions in one or two sentences:**(5x2=10Marks)*

- Q I**
- 1 Which are the various grades of concrete used for RCC?
 - 2 Distinguish between a beam and a Lintel.
 - 3 Which are the two ways of spanning a stair?
 - 4 What is meant by laterally supported beam?
 - 5 Write the equation and explain the notations for the design strength of a tension member in rupture.

PART - B*Answer any five from the given seven questions**(5x6=30Marks)*

- Q II**
- 1 Derive the equation for the Neutral axis depth of a singly reinforced section.
 - 2 Distinguish between singly and doubly reinforced beam and state the circumstances in providing a doubly reinforced beam.
 - 3 How the slabs are classified? How are they distinguished?
 - 4 When and where a combined footing is provided? What are the precautions to be taken in providing them?
 - 5 Determine the net area A_n for the tension member shown in figure when:
 (a) holes are drilled (b) holes are punched.



- 6 Which are the different failure modes of an axially loaded column?
- 7 Explain the functions with sketches of any four of the following parts of a truss:
 (a) top chord (b) bottom chord (c) sling (d) strut (e) sag tie.

PART – C (answer one full question from each module)

Module – I

- Q III** (a) List the various laboratory tests conducted on ingredients of concrete. 7
- (b) Find the ultimate resisting moment of the given beam of size 230x360mm effective depth with 4 numbers of 12mm dia steel using M20 and Fe 415 grade steel. 8

OR

- Q IV** (a) Which are different types of loading over a lintel? How are they calculated? 6
- (b) Find the limiting resisting moment of a flanged beam for the following details:
- | | |
|-------------------------------|------------------------|
| Grade of concrete-M20 | Grade of steel-Fe 415 |
| Breadth of web-250mm | Effective depth -510mm |
| Effective flange width-1000mm | Depth of flange-100mm |
- 9

Module – II

- Q v** (a) What is the function of distribution reinforcement in a slab? Which diameters of steel are used as distribution steel? 6
- (b) Design a one way slab for a clear span of 3m using M20 and Fe415 steel. The slab has a supporting wall of 230mm thickness and carries a super imposed load of 3kN/m^2 including finishes. 9

OR

- Q VI** (a) What are the standard values recommended for the following:
- (1) Minimum width of landing. (2) Range of rise (3) Range of tread.
(4) Inclination of stair (5) Minimum Head room (6) Height of handrail
(7) Number of risers in a flight. 7
- (b) Design a short column to carry an axial load of 1500kN using M25 concrete and 1.2% steel of Fe 415. 8

Module – III

- QVII** (a) What are the section parameters available from the steel table for an RSJ 6
- (b) Design a single angle tension member to carry a factored tensile force of 250kN. The member is subjected to possible reversal of stress due to wind action. The length of the member is 3m. Use 20mm shop bolts of grade 4.6 for the connection. Check the section only for yielding. 9

OR

- QVIII** (a) What are the merits and demerits of steel structures over an RCC 5

- (b) Design a longitudinal fillet weld to connect a plate of 100x12mm to a gusset plate of 160x12mm (Fe 250) to transmit a pull equal to the strength of the plate. The grade of weld is $f_u=410$ and welding is done in the workshop. 10

Module – IV

- QIX** (a) Differentiate a plated beam from a plate girder. 6
- (b) I beams provided in a hall has an effective span of 8.3m and spacing of 3m. The beam has an RCC slab of 100mm thickness, finishing load of 1.5 kN/m^2 , live load of 2 kN/m^2 and self weight of beam as 0.8 kN/m . Design a suitable section and check for shear. 9

OR

- QX** (a) What are the different types of sections used as purlins? What governs the spacing of purlins over a truss? 6
- (b) Design a single angle strut connected to the gusset plate to carry 180kN factored load. The length of the strut between centre to centre of intersection is 3m. 9